Question 6

a) $\hat{P}(A \mid X_1 = 40, X_2 = 3.5) = \sigma(\hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2) = \sigma(-6 + 0.05 \cdot 40 + 3.5) = \sigma(-0.5) \approx 0.378$

Answer: The probability of the student getting an A (based on our linear model) is roughly 37.8 %

b)
$$P(A) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2)}}$$

$$\Leftrightarrow \quad \frac{1}{P(A)} = 1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2)}$$

$$\Leftrightarrow \quad \frac{1}{P(A)} - 1 = e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2)} \qquad \mid \ln(x)$$

$$\ln\left(\frac{1}{P(A)} - 1\right) = -\beta_0 - \beta_1 x_1 - \beta_2 x_2 \qquad \mid + \beta_0 \mid + \beta_2 x_2$$

$$\ln\left(\frac{1}{P(A)} - 1\right) + \beta_0 + \beta_2 x_2 = -\beta_1 x_1 \qquad \mid : (-\beta_1)$$

$$x_1 = \frac{\ln\left(\frac{1}{P(A)} - 1\right) + \beta_0 + \beta_2 x_2}{-\beta_1} = \frac{\cancel{\ln(1)}^{0} - 6 + 3.5}{-0.05} = \frac{6 - 3.5}{0.05} = 50$$

Answer: In order for a student with a 3.5 GPA to have a probability of 50% (based on ou
linear model), he would need to study 50h.

Question 9

a) odds $= \omega = \dfrac{p}{1-p} \qquad \mid \cdot (1-p)$

$$\Leftrightarrow \quad \omega(1-p) = p$$
$$\Leftrightarrow \quad \omega - \omega p = p \qquad \mid + \omega p$$
$$\Leftrightarrow \quad \omega = p + \omega p$$
$$\Leftrightarrow \quad \omega = (1+\omega) p \qquad \mid : (1+\omega)$$
$$\Leftrightarrow \quad p = \frac{\omega}{1+\omega}$$

$\omega = 0.37$

$$p = \frac{0.37}{1 + 0.37} = 0.27$$

Answer: With odds of 0.37, about 27% of people will in fact default

b) $p = 16\%$

$$\omega = \frac{p}{1-p} = \frac{0.16}{1 - 0.16} \approx 0.138$$

Answer: An individual with defaulting probability has odds of about 0.138 of defaulting.

Question 12

a) $$\hat{P}_r(Y=\text{apple} \mid X=x) = 1-\hat{P}_r(Y=\text{orange} \mid X=x) = 1 - \frac{\exp(\hat\beta_0+\hat\beta_1 x)}{1+\exp(\hat\beta_0+\hat\beta_1 x)} = \frac{1}{1+\exp(\hat\beta_0+\hat\beta_1 x)}$$

$$\omega = \frac{\hat{P}_r(Y=\text{orange} \mid X=x)}{\hat{P}_r(Y=\text{apple} \mid X=x)} = \frac{\dfrac{\exp(\hat\beta_0+\hat\beta_1 x)}{\cancel{1+\exp(\hat\beta_0+\hat\beta_1 x)}}}{\dfrac{1}{\cancel{1+\exp(\hat\beta_0+\hat\beta_1 x)}}} = \frac{\exp(\hat\beta_0+\hat\beta_1 x)}{1} = \exp(\hat\beta_0+\hat\beta_1 x)$$

$$\log \omega = \log(\exp(\hat\beta_0+\hat\beta_1 x)) = \hat\beta_0+\hat\beta_1 x$$

Answer: $\hat\beta_0+\hat\beta_1 x$

b) $$\hat{P}_r(Y=\text{orange} \mid X=x) = \frac{\exp(\hat\alpha_{00}+\hat\alpha_{01} x)}{\exp(\hat\alpha_{00}+\alpha_{01} x) + \exp(\hat\alpha_{a0}+\hat\alpha_{a1} x)}$$

$$\hat{P}_r(Y=\text{apple} \mid X=x) = 1-\hat{P}_r(Y=\text{orange} \mid X=x) = 1 - \frac{\exp(\hat\alpha_{00}+\hat\alpha_{01} x)}{\exp(\hat\alpha_{00}+\alpha_{01} x) + \exp(\hat\alpha_{a0}+\hat\alpha_{a1} x)} = \frac{\exp(\hat\alpha_{a0}+\hat\alpha_{a1} x)}{\exp(\hat\alpha_{00}+\hat\alpha_{01} x) + \exp(\hat\alpha_{a0}+\hat\alpha_{a1} x)}$$

$$\omega = \frac{\hat{P}_r(Y=\text{orange} \mid X=x)}{\hat{P}_r(Y=\text{apple} \mid X=x)} = \frac{\dfrac{\exp(\hat\alpha_{00}+\hat\alpha_{01} x)}{\cancel{\exp(\hat\alpha_{00}+\alpha_{01} x) + \exp(\hat\alpha_{a0}+\hat\alpha_{a1} x)}}}{\dfrac{\exp(\hat\alpha_{a0}+\hat\alpha_{a1} x)}{\cancel{\exp(\hat\alpha_{00}+\hat\alpha_{01} x) + \exp(\hat\alpha_{a0}+\hat\alpha_{a1} x)}}} = \frac{\exp(\hat\alpha_{00}+\hat\alpha_{01} x)}{\exp(\hat\alpha_{a0}+\hat\alpha_{a1} x)}$$

$$\log \omega = \log\left(\frac{\exp(\hat\alpha_{00}+\hat\alpha_{01} x)}{\exp(\hat\alpha_{a0}+\hat\alpha_{a1} x)}\right) = \log(\exp(\hat\alpha_{00}+\hat\alpha_{01} x)) - \log(\exp(\hat\alpha_{a0}+\alpha_{a1} x)) = \hat\alpha_{00}+\hat\alpha_{01} x - \hat\alpha_{a0} - \hat\alpha_{a1} x$$

$$= (\hat\alpha_{01} - \alpha_{a1})x + \hat\alpha_{00} - \hat\alpha_{a1}$$

Answer: $(\hat\alpha_{01} - \alpha_{a1})x + \hat\alpha_{00} - \hat\alpha_{a1}$

c) Since both models are based on the same datasets, and the log odds of softmax coding is equivalent to the "normal" coding (see p. 145), we can derive the following equation:

$$\hat\beta_0+\hat\beta_1 x = (\hat\alpha_{01} - \hat\alpha_{a1})x + (\alpha_{00} - \alpha_{a0})$$

$$\Rightarrow \quad \hat\beta_1 = (\hat\alpha_{01} - \hat\alpha_{a1}) \quad \wedge \quad \hat\beta_0 = \hat\alpha_{00} - \hat\alpha_{a0}$$

$$\Leftrightarrow \quad -1 = \hat\alpha_{01} - \hat\alpha_{a1} \quad \wedge \quad 2 = \hat\alpha_{00} - \hat\alpha_{a0}$$

d) I am guessing there has been a typo and given is $\hat{\alpha}_{o0} = 1.2$, $\hat{\alpha}_{o1} = -2$, $\hat{\alpha}_{a0} = 3$, $\hat{\alpha}_{a1} = 0.6$
Similar to c) we know that:

$$\beta_0 = \hat{\alpha}_{o0} - \hat{\alpha}_{a0} = 1.2 - 3 = -1.8$$

$$\beta_1 = \hat{\alpha}_{o1} - \hat{\alpha}_{a1} = -2 - 0.6 = -2.6$$

e) If the probability of one class (say oranges) is > 0.5, then we classify the data as that.
Therefore if $\hat{P}(Y = \text{oranges} \mid X = x) > 0.5$, we classify as an orange.

$$\hat{P}(Y = \text{oranges} \mid X = x) > 0.5 \iff \omega > \frac{0.5}{1 - 0.5} = 1$$

$$\implies \log \omega > 0$$

Since the logits are equivalent in both models, we will classify the same 100% of the time.